Q: Explain the A* search algorithm in detail with a neat example. Discuss its working, heuristic function, admissibility, and advantages and limitations. (10 Marks)

1) Introduction :

→ A* is an informed search algorithm.
→ It is used to find the shortest path from a start node to a goal node in a graph.
→ It uses both the actual cost and a heuristic estimate.

2) Evaluation function :

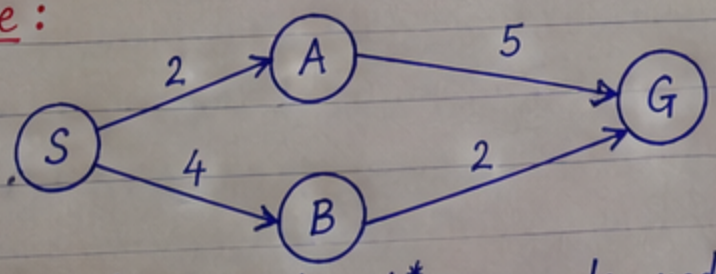
$$f(n) = g(n) + h(n)$$

→ $g(n)$: actual cost from start to node n.
→ $h(n)$: estimated cost from node n to goal.
→ $f(n)$: total estimated cost.

3) Working :

→ Keep an OPEN list and a CLOSED list.
→ Put the start node in OPEN list.
→ Select the node with the smallest $f(n)$.
→ If the node is goal, return the path.
→ Otherwise expand the node and add its successors.
→ Update $g(n)$, $h(n)$ and $f(n)$.
→ Repeat the process.

4) Example :



From the above graph, A* expands nodes based on $f(n)$ and finds the shortest path from S to G.